

Week 08 – Assignment

Scenario: Student Record Management

A college stores each student's information as a tuple because these details do not change during a semester.

One student's record is stored as:

("Ananya", 1023, "Computer Science", 8.7)

The information is stored in the following order:

- Student Name
- Student ID
- Department
- CGPA

The college uses this tuple whenever it needs to retrieve a student's information.

Q1. Which piece of information is stored in the third position of the tuple?

- A. Student Name
- B. Student ID
- C. Department
- D. CGPA

Correct Answer: C

Justification: The third value in the tuple is "Computer Science", which represents the student's department.

Q2. Why is a tuple a suitable data structure for storing this record?

- A. Because the student details are not expected to change frequently.
- B. Because tuples automatically sort the data.
- C. Because tuples allow duplicate student IDs.
- D. Because tuples can only store text values.

Correct Answer: A

Justification: Tuples are useful when the stored information remains fixed after creation.

Q3. How many pieces of information are stored in the student's tuple?

- A. 2

B. 3

C. 4

D. 5

Correct Answer: C

Justification: The tuple contains four values: Name, Student ID, Department, and CGPA.

Q4. Which of the following is the last item stored in the tuple?

A. Ananya

B. 1023

C. Computer Science

D. 8.7

Correct Answer: D

Justification: The last value in the tuple is the student's CGPA, which is 8.7.

Q5. A staff member wants to replace the department "Computer Science" with "Artificial Intelligence" in the existing tuple. What should they do?

A. Modify the existing tuple directly.

B. Create a new tuple with the updated value.

C. Delete only the department value.

D. Use the append() operation.

Correct Answer: B

Justification: Tuples are immutable, so their values cannot be modified after creation. A new tuple must be created with the updated information.

Scenario: Lucky Draw Betting Game

A gaming café has introduced a simple Lucky Draw Challenge. A player starts with ₹0 in their game account.

The game is played 7 times. In every round:

- The player's bet number is randomly chosen between 1 and 10.
- The lucky draw number is also randomly chosen between 1 and 10.
- If both numbers are the same, the player wins ₹900, but ₹100 is deducted as the participation fee. Hence, the account increases by ₹800.
- If the numbers are different, the player only loses the ₹100 participation fee.

- After every round, the current account balance is displayed.

The following Python program implements this game.

```
1. import random
2.
3. account = 0
4.
5. for i in range(7):
6.     bet = random.randint(1, 10)
7.     lucky_draw = random.randint(1, 10)
8.     print("Bet:", bet)
9.     print("Lucky draw:", lucky_draw)
10.
11.     if bet == lucky_draw:
12.         account = account + 900 - 100
13.     else:
14.         account = account - 100
15.
16.     print("Amount in your game account:", account)
```

Q1. According to Line 5, how many rounds will the game be played?

- A. 7
- B. 8
- C. 10
- D. Depends on the random numbers

Correct Answer: A

Justification: range(7) repeats the loop 7 times, so the game consists of seven rounds.

Q2. Which lines generate the random numbers for the player's bet and the lucky draw?

- A. Lines 3 and 5
- B. Lines 8 and 9
- C. Lines 11 and 12
- D. Lines 6 and 7

Correct Answer: D

Justification: Lines 6 and 7 use random.randint(1, 10) to generate the bet number and the lucky draw number.

Q3. If the bet number and lucky draw number are equal, what is the net change in the account balance according to Line 12?

- A. +₹100
- B. +₹900
- C. +₹700
- D. +₹800

Correct Answer: D

Justification: The player receives ₹900 and pays a ₹100 participation fee. Therefore, the net increase is ₹800.

Q4. According to Lines 13–14, what happens when the bet number and lucky draw number are different?

- A. ₹100 is deducted from the account.
- B. ₹900 is added to the account.
- C. The account balance remains unchanged.
- D. A new account is created.

Correct Answer: A

Justification: The else block subtracts ₹100 from the account, representing the participation fee.

Q5. (MSQ) Which of the following statements about the program are correct? (*Choose all that apply.*)

- A. The player's account balance is printed after every round.
- B. The player's bet is entered by the user using input().
- C. The lucky draw number is always different from the bet number.
- D. The player's account balance can become negative.

Correct Answers: A and D

Justification:

- A Line 16 prints the account balance after each iteration of the loop.
- B The program generates the bet randomly; it does not use input().
- C Since both values are random, they may or may not be equal.
- D If the player loses multiple rounds, ₹100 is deducted each time, so the account balance can become negative.

Scenario: Secure Access Puzzle Generator

A cybersecurity training platform uses anagrams to create authentication puzzles. Before granting access to a practice lab, the system displays a shuffled version of a secret keyword. To ensure the puzzle is meaningful, the displayed word must not be identical to the original keyword. The system keeps generating new arrangements until a valid anagram is produced. The following Python program implements this logic.

```
1. import random
2.
3. keyword = input("Enter the secret keyword: ")
4. letters = list(keyword)
5.
6. while True:
7.     random.shuffle(letters)
8.     candidate = "".join(letters)
9.
10.    if candidate != keyword:
11.        break
12.
13. print("Authentication Puzzle:", candidate)
```

Q1. Suppose the user enters the word LEVEL. Which of the following statements is always true?

- A. The displayed puzzle will never contain repeated letters.
- B. The displayed puzzle will always have exactly five letters.
- C. The displayed puzzle will always be alphabetically sorted.
- D. The displayed puzzle cannot be generated because the word contains repeated letters.

Correct Answer: B

Justification: Shuffling changes only the order of characters. The number of characters always remains the same. Repeated letters do not prevent an anagram from being generated.

Q2. Consider the execution of Lines 6–11. Which situation can cause the loop to execute more than once?

- A. The entered keyword has fewer than five letters.
- B. `random.shuffle()` arranges the letters in the same order as the original keyword.
- C. The keyword contains only uppercase letters.

D. The keyword contains duplicate characters.

Correct Answer: B

Justification: If the shuffled arrangement is identical to the original word, the condition on Line 10 fails and the loop repeats until a different arrangement is obtained.

Q3. Which of the following inputs may cause the program to never terminate?

A. PYTHON

B. RANDOM

C. A

D. SECURE

Correct Answer: C

Justification: A single-character word has only one possible arrangement. Therefore, candidate will always be equal to keyword, making the condition on Line 10 false forever.

Q4. Which statement best describes the purpose of Lines 10–11?

A. They ensure that every possible anagram is generated exactly once.

B. They ensure that duplicate letters are removed.

C. They stop the loop only after a shuffled word different from the original has been generated.

D. They guarantee that the generated anagram is alphabetically smallest.

Correct Answer: C

Justification: The loop terminates only when the generated arrangement differs from the original word.

Q5. (MSQ) Which of the following statements about the program are correct? (Choose all that apply.)

A. If the input contains only one character, the program may run indefinitely.

B. Every execution of the program produces a different anagram.

C. The program guarantees that all possible anagrams will eventually be generated.

D. The generated puzzle always contains exactly the same characters as the original keyword.

Correct Answers: A and D

Justification:

- A Correct. A one-character word has only one possible arrangement, so the loop never satisfies the condition on Line 10.

- B Incorrect. Randomness does not guarantee a different output on every execution. The same anagram may be generated in different runs.
- C Incorrect. The program stops after generating the first valid anagram. It does not enumerate all possible anagrams.
- D Correct. `random.shuffle()` only changes the order of the existing characters; it does not add, remove, or replace any characters.

Scenario: Customer Feedback Analysis

An online shopping company collects customer reviews after every purchase. The data is stored in a Pandas DataFrame. To understand customer satisfaction, the company uses NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analyzer. Each review is assigned a compound sentiment score. Reviews with a positive compound score are labeled Positive; otherwise, they are labeled Negative.

The following Python program performs this analysis.

```
1. import pandas as pd
2. from nltk.sentiment.vader import SentimentIntensityAnalyzer
3.
4. reviews = pd.DataFrame({
5.     "Customer": ["Asha", "Rahul", "Neha", "Arjun"],
6.     "Review": [
7.         "Excellent product and fast delivery",
8.         "The quality is average",
9.         "Very disappointed with the service",
10.        "Worth every penny"
11.    ]
12. })
13.
14. analyzer = SentimentIntensityAnalyzer()
15.
16. def get_sentiment(review):
17.     score = analyzer.polarity_scores(review)["compound"]
18.     if score > 0:
19.         return "Positive"
20.     else:
21.         return "Negative"
22.
23. reviews["Sentiment"] = reviews["Review"].apply(get_sentiment)
24.
25. print(reviews)
```

Q1. What is the primary purpose of Line 14?

A. To create a Pandas DataFrame.

- B. To create an object that can analyze the sentiment of text.
- C. To read customer reviews from a file.
- D. To display the final DataFrame.

Correct Answer: B

Justification: `SentimentIntensityAnalyzer()` creates a VADER sentiment analyzer that is used to calculate sentiment scores for text.

Q2. According to Line 17, which value is used to determine whether a review is positive or negative?

- A. neg
- B. compound
- C. neu
- D. pos

Correct Answer: B

Justification: The program extracts the compound score from the dictionary returned by `polarity_scores()`. This score is used in the decision on Lines 18–21.

Q3. What is the purpose of Line 23?

- A. It removes reviews with negative sentiment.
- B. It creates a new column containing the sentiment label for each review.
- C. It sorts the reviews based on sentiment.
- D. It replaces the original review text with sentiment scores.

Correct Answer: B

Justification: The `apply()` function calls `get_sentiment()` for every review and stores the returned labels in the new Sentiment column.

Q4. Which statement best describes the role of the function `get_sentiment()`?

- A. It converts reviews into uppercase.
- B. It counts the number of words in each review.
- C. It calculates the average sentiment score of all reviews.
- D. It assigns a sentiment label to a single review based on its compound score.

Correct Answer: D

Justification: The function computes the compound score for one review and returns either "Positive" or "Negative".

Q5. (MSQ) Which of the following statements are correct? (Choose all that apply.)

- A. The program stores the sentiment score directly in the Review column.
- B. The reviews are stored in a Pandas DataFrame.
- C. The sentiment label for each review is generated using the `get_sentiment()` function.
- D. The program classifies reviews based on the length of the review text.

Correct Answers: B and C

Justification:

- A. Incorrect. The original Review column is not modified.
- B. Correct. Lines 4–12 create a Pandas DataFrame named reviews.
- C. Correct. Line 23 applies the `get_sentiment()` function to every review.
- D. Incorrect. Classification is based on the compound sentiment score, not the length of the review.